CroCo-StableBridge: Risk-Aware Input Routing for a Frozen Stereo Matcher

Shao-Kai Liu
National Yang Ming Chiao Tung University

Yun-Tze Tsai
National Yang Ming Chiao Tung University

Chih-Chung Hsu
National Yang Ming Chiao Tung University
National Cheng Kung University

Abstract

Challenge entry: RoCo-22 · cactus8603. Simple preprocessing can improve stereo matching under image corruption, but a treatment that helps one pair may harm another. We present StableBridge, an input selector for a frozen CroCo-Stereo matcher. Early encoder features propose denoising or sharpening and estimate its benefit; deeper features estimate its harm. StableBridge applies the proposal only when the benefit score exceeds the harm score by a fixed margin and otherwise leaves the pair unchanged. In scene-held-out cross-validation on Spring with synthesized corruptions, it reduces corrupted-crop mean absolute disparity error (Abs) by 5.08% without changing clean-crop error. At matched total modification counts over clean and corrupted examples, the harm check lowers the fraction of changed corrupted crops that worsen by 4.55 percentage points, although it does not clearly improve overall gain under scene resampling. The official system obtains RbS 0.1456; without a matched CroCo-only submission, this is an end-to-end result rather than an estimate of the selector's incremental effect. Code is available at https://github.com/cactus8603/stablebridge-stereo.

1 Introduction

Before stereo matching, a stereo pair may be left unchanged or lightly processed by denoising or sharpening; we refer to these alternatives as treatments. Choosing among them is risky: denoising can suppress noise but may remove fine texture needed for matching, whereas sharpening may recover detail but also amplify artifacts. The best treatment can vary even within a single corruption type, depending on scene texture, corruption severity, and disparity range. A practical selector should therefore predict whether a treatment will help the matcher and retain the unchanged pair when uncertain. Figure 1 shows a clean pair for which denoising is harmful and a noisy pair for which it is beneficial.

RoCo-Spring evaluates clean Spring error [1] and penalizes disagreement between clean and corrupted predictions [2]. Robustness gains must therefore be weighed against any loss in clean accuracy. Our local analyses further show that no fixed treatment captures most of the gain available on individual crops and that a selector based only on image statistics can be unsafe on individual scenes. These observations motivate choosing a treatment for each stereo pair after estimating both benefit and harm. Section 4 defines the local and official measures separately: local supervision uses ground-truth disparity to measure treatment outcomes, whereas the official benchmark measures disagreement between predictions on clean and corrupted images.

40th Conference on Neural Information Processing Systems (NeurIPS 2026). Workshop: RoCo-Spring: The Robust Correspondence Challenge.

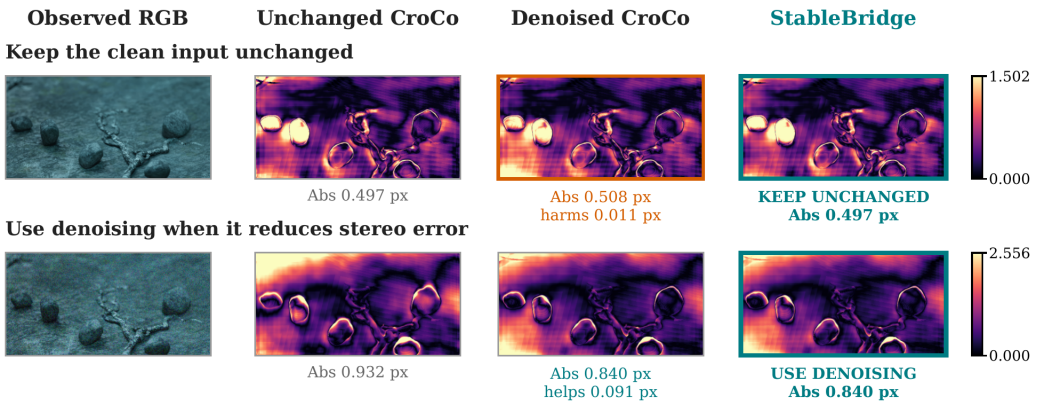


Figure 1: Denoising helps the Gaussian-noise example but harms its clean counterpart. On scene 0022, outside the 36 development scenes, clean Abs rises from 0.497 to 0.508 px, while noisy Abs falls from 0.932 to 0.840 px. Error maps are computed against ground-truth disparity and share a color scale within each row.

Contributions. We introduce a two-stage input selector for frozen stereo matching: early features propose a treatment, and an acceptance test combines its early benefit score with a harm score from deeper features. We evaluate treatment selection with scene-held-out tests and a matched-count comparison of the harm check. A five-crop diagnostic further examines whether the selected treatment remains useful across image regions.

"Bridge" denotes the connection from optional treatments to the frozen matcher; "Stable" denotes the identity fallback when the selected proposal fails the acceptance test.

2 Related Work

CroCo v2 [4] couples cross-view-completion pretraining with a DPT disparity head; we freeze its released Spring checkpoint. RobustSpring [2] extends Spring with 20 stereo-consistent corruption categories. RobuSTereo [5], Stereo Anywhere [6], FoundationStereo [7], and SMoEStereo [8] modify matcher training, representations, or internal adapters. Our selector operates before the released matcher and leaves its weights untouched.

Image processing may target reconstruction, as in PromptIR [12], or downstream utility. Dynamic image enhancement optimizes recognition performance [9]; UniRestore trains a task adapter with downstream losses [10]; and PRISM evaluates restored images with pretrained downstream models [11]. Most closely related, the concurrent TaskGuard method chooses between preserved and restored inputs for a frozen detector [13]. StableBridge makes this choice for stereo pairs and draws its scores from two encoder depths. Its accept-or-reject rule is related to selective prediction [14]. Unlike test-time adaptation [15, 16], StableBridge keeps the matcher fixed at inference. Its scores are trained on changes in disparity error, rather than perceptual quality as in no-reference IQA [17].

3 Method

Treatments and supervision. For a stereo pair $x_i = (I_i^L, I_i^R)$, disparity ground truth y_i, frozen matcher f_θ, and treatment T_k, define

$$L_{ik} = \ell(f_\theta(T_k(x_i)), y_i), \qquad g_{ik} = L_{i,\mathrm{id}} - L_{ik}. \tag{1}$$

Here L_{ik} is the mean absolute disparity error over valid pixels; positive g_{ik} means that T_k lowers the error relative to the unchanged input. The deployed system uses only $\mathcal{A}_3 = \{T_{\mathrm{id}}, T_{\mathrm{dn}}, T_{\mathrm{sh}}\}$: the identity treatment, a separable $[0.25, 0.5, 0.25]$ binomial filter (mild denoising), and unsharp masking (mild sharpening; radius 2, strength 150%, threshold 2). For analysis only, a larger set $\mathcal{A}_5$ also includes joint percentile normalization and chroma compression. Every non-identity treatment uses identical parameters in both views and does not change image geometry. The identity treatment passes decoded `uint8` RGB arrays to CroCo's standard normalization without treatment-side

modification. We fixed $\mathcal{A}_3$ before training because each added candidate requires its own CroCo feature pass and prediction-head outputs, followed by fresh training, threshold selection, and harm validation; Appendix A.1 gives the exact definitions and analysis-only results.

CroCo features. All three candidate pairs use the same frozen CroCo patch embedding and encoder. At depths $d \in \{1, 8\}$, a fixed module pools the left/right tokens and their differences from the unchanged input's tokens into a feature vector $z_{ik}^{(d)} \in \mathbb{R}^{227}$. Its 35 hand-designed summary features comprise 20 statistics of epipolar correlation along image rows, five of token change, four of spatial variation, five of asymmetry between left and right token norms, and one of global similarity between views. Another 192 features provide a label-free nonlinear summary from a fixed, randomly initialized patch MLP, avoiding a second feature-extractor fit within every scene fold on only 36 development scenes. This feature extractor has about 183,000 parameters, is initialized once before any labels are loaded, is shared across depths and folds, and is never trained; only the prediction heads are trained separately for each depth, seed, and cross-validation fold. Appendix A.2 gives the exact definitions.

Treatment proposal and harm check. StableBridge first proposes a treatment and then decides whether to apply it. At block 1, a head proposes whichever of denoising and sharpening has the larger conservative gain score and produces its benefit score. All three candidates then continue to block 8, whose head produces the proposed treatment's harm score; block 1 is not an early exit. The treatment is applied only when the early benefit score minus the deep harm score reaches a fixed threshold; otherwise, the pair remains unchanged. Writing $\widehat{r}$ for the benefit and harm scores after the sigmoid, the deployed rule is

$$\widehat{k}_i = \arg\max_{k \in \{\mathrm{dn},\mathrm{sh}\}} \widehat{q}_{.10,ik}, \quad s_i = \widehat{r}^{\,\mathrm{early}}_{\mathrm{benefit},i\widehat{k}_i} - \widehat{r}^{\,\mathrm{deep}}_{\mathrm{harm},i\widehat{k}_i}, \quad \pi_\tau(i) = \begin{cases} T_{\widehat{k}_i}, & s_i \geq \tau, \\ T_{\mathrm{id}}, & s_i < \tau. \end{cases} \tag{2}$$

At each depth, an MLP of width 128 shared across treatments receives the features for the candidate and unchanged pair, their difference, the elementwise sum of the mean and maximum over all three candidates, and a learned treatment embedding with eight dimensions. Each head produces a gain estimate trained with weighted Smooth-L1 loss, a conservative gain score $\widehat{q}_{.10}$ trained with weighted asymmetric (pinball) loss, and logits for benefit, harm, and severe harm. The deployed rule uses the conservative gain and benefit outputs from block 1 and the harm output from block 8. At inference, the conservative gain score ranks treatments, while the difference between the uncalibrated benefit and harm scores determines whether to apply the proposed treatment. The score for Abs increases above 0.05 px serves as an auxiliary diagnostic and is omitted from Eq. 2. We label a treatment harmful when it raises Abs by more than 0.001 px and severely harmful when the increase exceeds 0.05 px; evaluation reports the corresponding harm rates among changed corrupted crops.

Each depth has a separate head trained with the same weighted objective,

$$\begin{aligned} \mathcal{L} = {} & 2\mathcal{L}_{\text{Smooth-L1}} + 0.75\mathcal{L}_{q=.1} + 0.75\mathcal{L}_{\text{benefit}} + 0.75\mathcal{L}_{\text{harm}} \\ & + \mathcal{L}_{\text{severe}} + \mathcal{L}_{\text{rank}} + 0.25\mathcal{L}_{\text{order}}. \end{aligned} \tag{3}$$

The two regression terms fit gain and its conservative score; three weighted binary losses train benefit, harm, and severe harm; and the final terms train the treatment ranking and penalize conservative scores that exceed the gain estimate. Appendix A.2 gives every target, weight, and formula.

Deployment at full resolution. Five crops (center and four quadrants) vote on each complete 1080×1920 pair. Denoising or sharpening is applied to both views only if at least three crops select that same treatment; otherwise the pair stays unchanged. Frozen CroCo merges disparities inferred with the left and right images as references in overlapping windows to produce the official output at full resolution (Appendix A.5).

Figure 2 summarizes the decision for each crop and the final choice for the stereo pair.

4 Experimental Protocol

Local evaluation set. From 144 Spring training pairs in 36 scenes, we take a fixed 352×704 center crop from four frames per scene, each clean and under 20 synthesized corruptions at three numbered

severity levels (1–3). This gives 8,784 examples and 26,352 outcomes for the deployed treatments. After sorting the N frames in each scene by filename, we use indices $\lfloor jN/5 \rfloor$, $j = 1, \ldots, 4$, giving equal examples per scene and limiting the analysis with five treatments to 43,920 CroCo passes. We fixed this rule and base seeds 17, 43, and 89 before nested evaluation. The selector receives no corruption label, severity, scene identifier, GT, or final CroCo prediction (Appendix A.2).

Local metrics and statistics. To compare selection rules π on a common scale, we report the fraction of the GT gain from the best of five treatments achieved by each rule:

$$C_5(\pi) = \frac{\sum_{i \in \mathcal{C}} g_{i,\pi(i)}}{\sum_{i \in \mathcal{C}} \max_{k \in \mathcal{A}_5} g_{ik}}. \tag{4}$$

Here $\mathcal{C}$ contains equally weighted corrupted crops. In the denominator, GT selects the treatment in $\mathcal{A}_5$ with the lowest error for each crop; including the unchanged option prevents negative gain, and the upper bound is 100% by construction. This GT-dependent reference is used only as an analysis upper bound.

We also report overall Abs gain, mean Abs gain across corruption types, the fraction of valid pixels with disparity error above 1 px, D1 (the fraction with error above both 3 px and 5% of the GT disparity magnitude), and modification and harm rates. GT supplies the training targets and scores local outcomes, but scene-held-out predictions and treatment decisions are produced without GT before those outcomes are evaluated. For center crops, each example contributes its mean error with equal weight. For evaluation on five crops, absolute errors are pooled over all valid pixels in the scored crops. We bootstrap complete scenes 5,000 times and report the 2.5th and 97.5th percentiles of the resulting metric estimates. Because models, thresholds, and decisions remain fixed, these ranges measure sensitivity to scene composition and exclude retraining and model-selection variation (Appendix A.3).

Nested cross-validation over scenes. We split the 36 scenes into six groups of six. Each *outer fold* holds out one group as test scenes, which are used neither to train the heads nor to choose τ. The remaining 30 scenes form five *inner folds*: each trains heads on 24 scenes and scores the other six, so every one of the 30 scenes is scored by a model that never saw it. On these pooled scores we test up to 401 thresholds, keep those with clean gain $\geq -0.2\%$, gain ≥ 0 in each corruption family (color, blur, noise, quality, weather; Table 4 in Appendix A.3), and worst-scene gain $\geq -0.5\%$, and pick the highest C_5 (tie-breaks in Appendix A.3). Heads retrained on all 30 scenes are then evaluated once on the held-out group at that τ. We call such predictions *scene-held-out*; Table 5 in Appendix A.3 lists each scene's role at every stage.

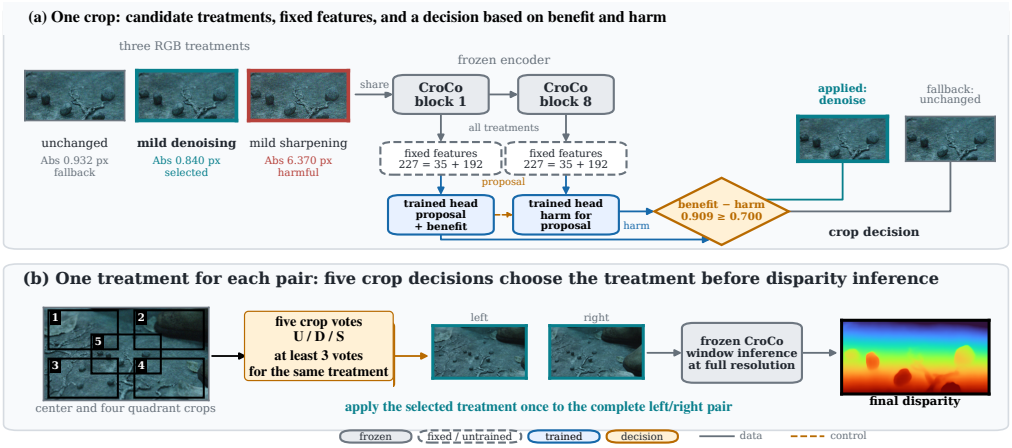


Figure 2: StableBridge selects one treatment before frozen CroCo. (a) For the Gaussian-noise crop from scene 0022, 227-dimensional fixed features feed the block-1 proposal and benefit outputs and the block-8 harm output. Denoising passes the fixed-margin test; a failed test would retain the unchanged pair. (b) The pair adopts a treatment when at least three of five crops agree. Displayed Abs values are computed against ground-truth disparity.

Applying the same rule to all 8,784 scene-held-out scores gives the deployment threshold $\tau = 0.700$ (the exact value is given in Appendix A.3), after which six heads are retrained on all 36 scenes. The heads, feature extractor, and threshold are frozen before evaluating scene 0022 or running official inference. The development scenes also informed method design, so scene-held-out results do not independently evaluate architecture selection.

Training. Each head trains for 80 epochs with AdamW (batch size 512, learning rate and weight decay 10^{-3}, cosine decay, gradient-norm clipping at 5). Seeds 17, 43, and 89 were fixed before evaluation, and the raw outputs of the three models are averaged to reduce dependence on a single initialization (Appendix B.2 reports every seed subset). Six outer folds $\times$ (five inner fits + one outer fit) $\times$ two depths $\times$ three seeds give 216 head fits for nested evaluation. The ensemble stores 992,181 selector parameters outside frozen CroCo (Appendix A.2).

Evaluating one treatment for each pair on five crops. After applying the voted treatment to each complete stereo pair, we run CroCo independently on the same five 352×704 crops and score disparity predictions that use the left image as reference. The 144 pairs are evaluated clean and at the middle severity (level 2 of 3) of each of the 20 corruptions (3,024 examples). The crops cover 55.65% of the image; pixels in overlapping regions are counted separately in each crop. This is a crop-based diagnostic; the official submission instead scores a complete disparity map at full resolution (Appendix A.5).

Official benchmark metric. The official RoCo-Spring score combines clean Spring error [1] with the RobustSpring disagreement between predictions on clean and corrupted images [2]:

$$\mathrm{RbS} = 0.5\frac{E_S}{B_S} + 0.5\frac{E_S + \Delta}{B_{R,\mathrm{proxy}}}, \tag{5}$$

For the submitted system, write $\widehat{d}_{\mathrm{SB}}(x) = F_\theta(\Pi_\tau(x))$, where Π_τ performs five-crop voting and applies the selected treatment to the complete pair, and F_θ is frozen CroCo's full-resolution window inference and merging pipeline (Appendix A.5). For clean x and corrupted x^c, define $\delta(x, x^c) = M_{\mathrm{Abs}}(\widehat{d}_{\mathrm{SB}}(x), \widehat{d}_{\mathrm{SB}}(x^c))$, the mean absolute difference over benchmark-scored pixels; Δ is its benchmark aggregate. E_S is clean Abs; $B_S = 3.4545$ and $B_{R,\mathrm{proxy}} = 18.4505$ are fixed normalization baselines [3]. Clean error contributes to both equally weighted components. All choices were frozen before official evaluation; local GT error and official prediction disagreement remain distinct.

5 Results

Crop-level performance. Table 1 summarizes the scene-held-out crop results. Across 8,640 corrupted crops, StableBridge lowers Abs from 2.724 to 2.586 px, a 5.08% reduction, while clean Abs remains 0.684 px. This reduction remains positive under scene resampling (3.60%–6.57%), and local disagreement between predictions on clean and corrupted images falls from 2.396 to 2.248 px. Color gain is -0.02% (Appendix A.3).

Table 1: Cross-validated crop results. Abs is measured on corrupted crops, and C_5 reports the fraction of the best-of-five GT gain achieved (Eq. 4; the three-treatment upper bound is 80.25%). Mean by type, Clean, and Worst scene are relative Abs gains. Harm is the harm rate among changed corrupted crops at each error-increase cutoff.

Rule	Abs	C_5	Mean by type	Clean	$\mathrm{Harm}_{.001}$	$\mathrm{Harm}_{.05}$	Worst scene
Unchanged CroCo	2.724	0.0%	0.00%	0.00%	—	—	0.00%
StableBridge	2.586	41.50%	+4.29%	**0.00%**	**22.04%**	**11.02%**	+1.02%
Rule using benefit only	2.581	42.86%	+4.37%	−0.38%	26.59%	12.30%	+1.11%
Selector based on image statistics	2.595	38.8%	+4.10%	−0.57%	37.4%	12.63%	−11.2%

With the block-1 proposal fixed, StableBridge and the benefit-only rule match total modification counts across clean and corrupted examples in each outer fold. The fraction of modified corrupted crops whose Abs increases by more than 0.001 px falls from 26.59% to 22.04%.

The paired 95% scene-bootstrap interval for this reduction is 2.67–6.39 percentage points. The C_5 difference (StableBridge minus benefit only) is -1.36 percentage points (95% interval $[-3.77, 1.45]$); neither overall gain nor C_5 shows a clear advantage.

The best fixed treatment captures 15.42% of the best-of-five GT gain; GT choices by corruption type and by type and severity reach 53.17% and 57.20%, and the three-treatment upper bound is 80.25%. StableBridge reaches 41.50%. The image-statistics selector reaches 38.76% but raises clean error by 0.57% and worst-scene error by 11.23% (Figure 3 in Appendix B.1).

Ablations. Under the same six-fold protocol, the full 227 features, the 35 summary statistics, and the 192 fixed-MLP features yield C_5 values of 41.50%, 41.18%, and 15.87%, respectively. Scene resampling finds no clear difference between the full and 35-feature models and favors both over the fixed-MLP-only model (Table 3 in Appendix A.2). On the development data, block 1 correlates more strongly with treatment gain than block 8 (0.551 versus 0.538), whereas block 8 has higher severe-harm AUROC (0.797 versus 0.745; Appendix B.7). These diagnostics use the 0.05-px severe-harm target; the deployed harm rate uses a 0.001-px cutoff. Eight-frame refitting did not improve mean gain, and no smaller seed subset was consistently better under scene resampling (Appendix B.2). A fixed SwinIR JPEG restorer gave a 3.72% all-scene JPEG gain, but its range under scene resampling crossed zero and it increased error on clean and non-JPEG inputs; it therefore did not meet the prespecified condition for training a four-treatment selector (Appendix B.3).

Pair-level and official evaluation. The five-crop vote lowers Abs from 3.594 to 3.423 px (4.78%) and D1 from 11.77% to 11.25%, but the gains are concentrated in noise corruptions. The four noise types contribute about 90% of net gain across corruption types; the other 16 average $+0.42\%$. Speckle noise exposes a spatial failure: denoising improves the center crop by 18.16% while increasing error in the four quadrants by 2.64%, resulting in a net regression of 0.46% over the five crops (Figure 4 and Table 11 in Appendix B.5).

On the official benchmark, the five-crop vote leaves all 1,000 clean pairs unchanged and reproduces CroCo-Stereo's released 1px/Abs/D1 scores of 7.135%/0.471 px/2.705%. The server reports 22.529%/2.386 px/4.427% disagreement on corrupted pairs, proxy Abs 2.857 px, and RbS 0.1456. The required 21,000-pair matched CroCo-only inference and validation could not be allocated within the revision compute budget, so a matched server run was not completed. These numbers therefore describe the submitted system end to end and do not isolate StableBridge's incremental effect; Appendix B.6 gives the treatment counts.

6 Limitations and Conclusion

Limitations. Results cover one matcher and dataset, with local tests on crops rather than complete disparity maps. The local renderer approximates motion blur and the five weather effects (fog, frost, rain, snow, and spatter); motion blur uses an image-space line kernel. The 36 development scenes also informed method design, so scene-held-out results do not independently evaluate architecture selection. Most gain comes from noise: pair-level selection gains at most 1.53% on the four corruptions with the largest official disagreement. The learned-restorer check covered only SwinIR for JPEG. Some center crops are harmed, and pair-level selection raises worst-scene error by 4.57%. Without a matched official unchanged-CroCo run, the reported RbS score cannot quantify StableBridge's incremental contribution on the official benchmark.

Conclusion. StableBridge uses benefit and harm scores to gate denoising or sharpening before frozen CroCo. In this evaluation, pair-specific routing lowers corrupted-crop error without changing clean-crop error, and the matched-count comparison shows a lower harm rate but no clear overall-gain advantage. The gains do not transfer uniformly across image regions or corruption types; the weak results on blur and weather point to the next practical step: a broader treatment set.

References

[1] L. Mehl, J. Schmalfuss, A. Jahedi, Y. Nalivayko, and A. Bruhn. Spring: A high-resolution high-detail dataset and benchmark for scene flow, optical flow and stereo. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 4981–4991, 2023.

[2] V. Oei, J. Schmalfuss, L. Mehl, M. Bartsch, S. Agnihotri, M. Keuper, A. Bulling, and A. Bruhn. RobustSpring: Benchmarking robustness to image corruptions for optical flow, scene flow and stereo. In *International Conference on Learning Representations (ICLR)*, 2026.

[3] RoCo-Spring Organizers. RoCo-Spring: Rankings, metrics & submissions. https://roco-spring.github.io/evaluation.html, 2026. Accessed September 14, 2026. Stereo Matching scoring convention.

[4] P. Weinzaepfel, T. Lucas, V. Leroy, Y. Cabon, V. Arora, R. Brégier, G. Csurka, L. Antsfeld, B. Chidlovskii, and J. Revaud. CroCo v2: Improved cross-view completion pre-training for stereo matching and optical flow. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 17969–17980, 2023.

[5] Y. Wang, Y. Liang, Y. Hu, and Y. Fu. RobuSTereo: Robust zero-shot stereo matching under adverse weather. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 25134–25144, 2025.

[6] L. Bartolomei, F. Tosi, M. Poggi, and S. Mattoccia. Stereo anywhere: Robust zero-shot deep stereo matching even where either stereo or mono fail. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 1013–1027, 2025.

[7] B. Wen, M. Trepte, J. Aribido, J. Kautz, O. Gallo, and S. Birchfield. FoundationStereo: Zero-shot stereo matching. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 5249–5260, 2025.

[8] Y. Wang, L. Wang, C. Zhang, Y. Zhang, Z. Zhang, A. Ma, C. Fan, T. L. Lam, and J. Hu. Learning robust stereo matching in the wild with selective mixture-of-experts. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 21276–21287, 2025.

[9] Vivek Sharma, Ali Diba, Davy Neven, Michael S. Brown, Luc Van Gool, and Rainer Stiefelhagen. Classification-driven dynamic image enhancement. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 4033–4041, 2018.

[10] I-Hsiang Chen, Wei-Ting Chen, Yu-Wei Liu, Yuan-Chun Chiang, Sy-Yen Kuo, and Ming-Hsuan Yang. UniRestore: Unified perceptual and task-oriented image restoration model using diffusion prior. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 17969–17979, 2025.

[11] Rupa Kurinchi-Vendhan, Pratyusha Sharma, Antonio Torralba, and Sara Beery. Seeing through the PRISM: Compound & controllable restoration of scientific images. In *International Conference on Learning Representations (ICLR)*, 2026.

[12] V. Potlapalli, S. W. Zamir, S. Khan, and F. S. Khan. PromptIR: Prompting for all-in-one image restoration. In *Advances in Neural Information Processing Systems*, volume 36, pages 71275–71293, 2023.

[13] Vung Pham. TaskGuard: Task-conditioned restoration utility for risk-aware object detection. *arXiv preprint arXiv:2609.08011*, 2026.

[14] Yonatan Geifman and Ran El-Yaniv. SelectiveNet: A deep neural network with an integrated reject option. In *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, pages 2151–2159, 2019.

[15] Dequan Wang, Evan Shelhamer, Shaoteng Liu, Bruno Olshausen, and Trevor Darrell. Tent: Fully test-time adaptation by entropy minimization. In *International Conference on Learning Representations (ICLR)*, 2021.

[16] Qin Wang, Olga Fink, Luc Van Gool, and Dengxin Dai. Continual test-time domain adaptation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 7201–7211, 2022.

[17] Hossein Talebi and Peyman Milanfar. NIMA: Neural image assessment. *IEEE Transactions on Image Processing*, 27(8):3998–4011, 2018.

[18] Jingyun Liang, Jiezhang Cao, Guolei Sun, Kai Zhang, Luc Van Gool, and Radu Timofte. SwinIR: Image restoration using Swin transformer. In *Proceedings of the IEEE/CVF International Conference on Computer Vision Workshops (ICCVW)*, pages 1833–1844, 2021. doi: 10.1109/ICCVW54120.2021.00210.

A Reproducibility details

A.1 Exact treatment definitions

The analysis set contains five deterministic treatments that preserve geometry; the deployed selector uses the first three. The outputs of all floating point image operations are rounded and converted back to `uint8` before CroCo normalization, and the same parameters are applied to both views; only two treatments estimate their parameters jointly from pixels in both views. The identity treatment copies the decoded left/right `uint8` RGB arrays without filtering, color conversion, resampling, or re-encoding before normalization. Here "bit-exact unchanged input" refers to these decoded arrays before normalization; compressed-file bytes and normalized floating point tensors lie outside the definition. In the local set of 8,784 examples and in deployment, treatments are applied to the complete `uint8` views before selector crops are extracted, so crop edges do not invoke filter padding.

Table 2: Fixed treatments. The two "analysis only" treatments enter the comparison whose upper bound uses all five treatments but are unavailable to the deployed selector.

Role	Name	Definition
deployed	unchanged	Decoded `uint8` RGB arrays copied without pixel modification
deployed	mild denoising	OpenCV 4.13.0 separable $[0.25, 0.5, 0.25]$ kernel on `float32`, reflect-101 boundary, nearest-integer rounding
deployed	mild sharpening	Pillow 12.2.0 `UnsharpMask(radius=2.0, percent=150, threshold=2)`, edge-replicating boundary
analysis only	percentile normalization	1st–99th percentiles pooled from both views, separately for each channel; minimum 8-bit range 12
analysis only	chroma compression	HSV saturation scaling from the 95th percentile pooled from both views; target 160, scale clipped to $[0.35, 1]$

For denoising, OpenCV's `sepFilter2D` applies the stated kernel along both axes to each `float32` RGB view; the result is rounded with `numpy.rint`, clipped to $[0, 255]$, and cast to `uint8`. Sharpening applies Pillow's stated filter independently to the two `uint8` RGB views. For percentile normalization, each RGB channel pools both views to obtain $l = q_{.01}$ and $h = q_{.99}$, sets $\beta = 255/\max(h - l, 12)$, and maps each view as $I' = \mathrm{uint8}(\mathrm{clip}(\mathrm{rint}((I - l)\beta), 0, 255))$. For chroma compression, both views are converted with OpenCV from RGB to HSV, and their saturation samples are pooled. If $q_{.95}$ is the joint 95th percentile, then

$$\alpha = \mathrm{clip}(160/\max(q_{.95}, 1), 0.35, 1)\,, \qquad S' = \mathrm{uint8}(\mathrm{clip}(\mathrm{rint}(\alpha S), 0, 255))\,.$$

The modified HSV views are converted back to RGB. Thus, 160 sets the scale at the joint 95th percentile; individual saturation values are multiplied by α, not interpolated toward 160.

The upper bound that chooses among $\mathcal{A}_3$ for each crop has $C_5 = 80.25\%$ and improves overall corrupted Abs by 9.821%. We fixed these three treatments before training the heads because every additional treatment requires its own CroCo feature pass and outputs from the prediction head. When applied to every pair, percentile normalization and chroma compression increase clean Abs by 1.71% and 0.20%, respectively. A selective version would require fresh training, threshold selection, and a harm check; that setting remains untested. Appendix B.3 separately tests one fixed pretrained SwinIR JPEG restorer and explains why it was not added.

A.2 Fixed CroCo features and trained prediction heads

Each candidate stereo pair is normalized with the released CroCo recipe, processed by CroCo's patch embedding, and encoded with frozen weights. The fixed feature extractor normalizes tokens with 1,024 dimensions, projects them to 96 dimensions, and uses their 22×44 spatial grid. Its 227 features comprise: 20 statistics of epipolar correlation (five summaries each of peak correlation, the gap between the two highest correlations, entropy, and expected horizontal offset over 0–32 token offsets); five statistics of the cosine change from the unchanged input; four statistics of spatial variation; five statistics of asymmetry between left and right token norms; one value for global similarity between views; and 192 features formed from 64 channels of a fixed patch MLP using mean pooling, maximum pooling, and attention-weighted pooling computed by the fixed, untrained module. The

five summaries are the mean, standard deviation, and the 10th, 50th, and 90th percentiles. Correlation summaries exclude columns with fewer than five valid offsets. Asymmetry between view norms is measured at equal token coordinates; no test of reciprocal disparity is computed.

Here is the exact computation. Let $t^v_{k,p} \in \mathbb{R}^{1024}$ be the frozen CroCo token for treatment k, view $v \in \{L, R\}$, and patch $p = (r, w)$. LayerNorm, a bias-free $1024 \to 96$ linear projection, and ℓ_2 normalization give $\widetilde{t}^v_{k,p}$. For a patch in a column with at least five valid offsets, let $D_w = \{0, \dots, \min(32, w)\}$ and define

$$c_{k,p,\delta} = (\widetilde{t}^L_{k,r,w})^\top \widetilde{t}^R_{k,r,w-\delta}, \qquad q_{k,p,\delta} = \frac{\exp(c_{k,p,\delta})}{\sum_{j \in D_w} \exp(c_{k,p,j})},$$

$$e_{k,p} = -\frac{\sum_{\delta \in D_w} q_{k,p,\delta} \log q_{k,p,\delta}}{\log |D_w|}, \qquad \bar{\delta}_{k,p} = \frac{1}{32} \sum_{\delta \in D_w} \delta q_{k,p,\delta}.$$

For each patch we take the largest c, the gap between its two largest values, e, and $\bar{\delta}$; taking five summaries of each quantity gives 20 features. The remaining 15 explicitly defined features are: five summaries of $1 - (\widetilde{t}^v_{k,p})^\top \widetilde{t}^v_{\mathrm{id},p}$ over both views and all patches; four features for spatial variation; five summaries of

$$\frac{\left| \|t^L_{k,p}\|_2 - \|t^R_{k,p}\|_2 \right|}{\|t^L_{k,p}\|_2 + \|t^R_{k,p}\|_2 + 10^{-6}};$$

and one value for similarity between views, $(\mathrm{mean}_p\, \widetilde{t}^L_{k,p})^\top (\mathrm{mean}_p\, \widetilde{t}^R_{k,p})$. The four features for spatial variation use $\widetilde{t}$ after LayerNorm, projection to 96 dimensions, and ℓ_2 normalization: horizontal and vertical mean squared coordinatewise differences are averaged over both views, all valid adjacent grid edges, and all 96 channels, and each mean is also differenced from its value for the identity treatment.

For the 192 fixed random features, let

$$\begin{aligned} x_{k,p} = [&\widetilde{t}^L_{k,p}, \widetilde{t}^R_{k,p}, |\widetilde{t}^L_{k,p} - \widetilde{t}^R_{k,p}|, \widetilde{t}^L_{k,p} - \widetilde{t}^L_{\mathrm{id},p}, \\ &\widetilde{t}^R_{k,p} - \widetilde{t}^R_{\mathrm{id},p}, \widetilde{t}^L_{k,p} \odot \widetilde{t}^R_{k,p}] \in \mathbb{R}^{576}, \\ h_{k,p} = &\ \mathrm{SiLU}(W_2\, \mathrm{SiLU}(W_1 x_{k,p} + b_1) + b_2) \in \mathbb{R}^{64}, \\ \alpha_{k,p} = &\ \mathrm{softmax}_p(w^\top h_{k,p} + b). \end{aligned}$$

The two linear layers are $576 \to 128 \to 64$; the attention layer is $64 \to 1$. Concatenating $\mathrm{mean}_p\, h_{k,p}$, the coordinatewise $\max_p h_{k,p}$, and $\sum_p \alpha_{k,p} h_{k,p}$ gives 192 features. Together with the 35 definitions above, this yields 227 exactly.

The module's 182,529 parameters are initialized once with seed 20260824 and never optimized. PyTorch's default Linear and LayerNorm initializers are used. The same fixed extractor is shared across both depths, every fold, and all three training seeds; its seed was fixed before any treatment outcome was loaded. We chose this label-free nonlinear summary to keep fitting tractable on 36 scenes. We did not compare trained projections, alternative fixed projections, or other projection seeds. A learned projection would also have to be fitted separately inside every fold to preserve the scene split.

Table 3: Feature-source ablation under the same six-fold nested protocol; all entries are percentages. $R(\mathcal{C})$ is the overall relative reduction in Abs on corrupted examples; Mean by type and Clean are relative Abs gains, Modified is the percentage assigned a treatment, and Harm/modified counts changed corrupted examples whose Abs rises by more than 0.001 px. Omitted branches are set to zero, preserving the 227-dimensional feature vector, the 916-value head input, and the parameter count.

Features	$R(\mathcal{C})$	Mean by type	C_5	Modified	Harm/modified	Clean
35 summary statistics + 192 fixed MLP	5.079	4.292	41.501	30.880	22.039	0.000
35 summary statistics only	5.039	4.277	41.180	27.245	22.387	−0.007
192 fixed MLP only	1.942	1.485	15.866	11.655	20.755	0.000

The two ablated variants reuse cached CroCo features from blocks 1 and 8 but repeat prediction-head fitting and inner-fold threshold selection. Under scene resampling, the 35-feature model and

the full model overlap in overall gain, mean gain across corruption types, and C_5; the 192-feature model alone is weaker on all three. We retain the submitted architecture, although this experiment finds no measurable gain from adding the fixed-MLP branch to the 35 summary statistics.

For treatment k, each prediction head concatenates (1) the candidate features, (2) the features for the identity treatment, (3) their difference, (4) the elementwise sum of the mean and maximum across treatments, and (5) a learned embedding $e_k \in \mathbb{R}^8$ that identifies the treatment. The input therefore has $4(227) + 8 = 916$ values. The exact trainable prediction head is

$$916 \xrightarrow{\text{Linear}} 128 \xrightarrow{\text{LayerNorm+SiLU}} 128 \xrightarrow{\text{Linear+SiLU}} 128 \xrightarrow{\text{Linear}} 6.$$

Each depth has a separate MLP that is shared across treatments, with width 128, LayerNorm, SiLU activations, and five trained outputs: a gain estimate trained with a weighted Smooth-L1 loss, a conservative gain score trained with asymmetric loss, and binary scores for benefit, harm, and severe harm. The checkpoint fields `mean_utility` and `q10_utility` name the Smooth-L1- and pinball-trained outputs. The sixth output, `support_logit`, has no training target or loss term and is ignored by treatment selection and evaluation; it is retained so that the prediction head matches the six-channel checkpoint interface. Equation 2 uses three of the five trained outputs. The learned embedding and all layers in the prediction head are fitted independently for every depth, fold, and seed.

The following definitions reproduce every term of the training loss. For crop example i and denoising or sharpening treatment $k \in \{\mathrm{dn}, \mathrm{sh}\}$, we normalize gain by the error on the unchanged image and define three binary targets:

$$u_{ik} = \mathrm{clip}\left(\frac{g_{ik}}{\max(L_{i,\mathrm{id}}, 10^{-5})}, -1, 1\right),$$
$$b_{ik} = \mathbf{1}[g_{ik} > 0.001], \quad h_{ik} = \mathbf{1}[g_{ik} < -0.001], \quad c_{ik} = \mathbf{1}[g_{ik} < -0.05].$$

All weights below are computed only from the scenes used to train the current model. Let $v_i = \max(0, g_{i,\mathrm{dn}}, g_{i,\mathrm{sh}})$. The typical positive gain m and example weight w_i are

$$m = \begin{cases} \mathrm{median}\{v_i : v_i > 0\}, & \text{if any } v_i > 0, \\ 1, & \text{otherwise}, \end{cases} \qquad w_i = \mathrm{clip}\left(1 + \frac{v_i}{\max(m, 10^{-6})}, 1, 12\right).$$

For each binary target $t \in \{b, h, c\}$ and denoising or sharpening treatment k, let N^+_{tk} and N^-_{tk} be its positive and negative training counts. The weight for the positive class is $\alpha_{tk} = \mathrm{clip}(N^-_{tk}/\max(N^+_{tk}, 1), 0.25, 12)$. For the gain losses, $a_{ik} = w_i(1 + 4|u_{ik}|)$; the factor $1 + 4|u_{ik}|$ equals 1 when the normalized gain u_{ik} is zero and gives outcomes with larger $|u_{ik}|$ more influence. With $\beta = 0.02$, the elementwise Smooth-L1 function is

$$H_\beta(e) = \begin{cases} e^2/(2\beta), & |e| < \beta, \\ |e| - \beta/2, & |e| \geq \beta. \end{cases}$$

$$\mathcal{L}_{\text{Smooth-L1}} = \mathrm{mean}_{i,k}[a_{ik} H_{.02}(\widehat{u}_{ik} - u_{ik})],$$
$$\mathcal{L}_{q=.1} = \mathrm{mean}_{i,k}[a_{ik} \max(0.1 r_{ik}, -0.9 r_{ik})], \quad r_{ik} = u_{ik} - \widehat{q}_{.10,ik}.$$

The weighted pinball-trained score is used for conservative treatment ranking and is not assumed to be a calibrated conditional quantile. Let z be a raw score before the sigmoid, $y \in \{0, 1\}$ its target, and α the weight for its positive class. Define $B(z, y; \alpha) = -\alpha y \log \sigma(z) - (1 - y) \log(1 - \sigma(z))$. Then $\mathcal{L}_t = \mathrm{mean}_{i,k}[w_i B(z^t_{ik}, t_{ik}; \alpha_{tk})]$ for $t \in \{b, h, c\}$. We denote these three losses by $\mathcal{L}_{\text{benefit}}$, $\mathcal{L}_{\text{harm}}$, and $\mathcal{L}_{\text{severe}}$. Let $\Delta u_i = u_{i,\mathrm{dn}} - u_{i,\mathrm{sh}}$ and $\rho_i = w_i \min(|\Delta u_i|, 1)$. The final two terms are

$$\mathcal{L}_{\text{rank}} = \frac{\sum_i \rho_i \,\mathrm{softplus}[-\mathrm{sign}(\Delta u_i)(\widehat{u}_{i,\mathrm{dn}} - \widehat{u}_{i,\mathrm{sh}})]}{\max(\sum_i \rho_i, 10^{-6})},$$
$$\mathcal{L}_{\text{order}} = \mathrm{mean}_{i,k}[\max(\widehat{q}_{.10,ik} - \widehat{u}_{ik}, 0)].$$

Equation 3 combines the seven loss terms. The ranking term trains $\widehat{u}$, while at test time treatments are ranked with the separately trained output that predicts conservative gain. The order term is an unweighted mean.

Prediction heads use AdamW for 80 fixed epochs, batch size 512, learning rate 10^{-3}, weight decay 10^{-3}, cosine decay of the learning rate, and a gradient norm clipped at 5. We retain the AdamW defaults $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$, and `amsgrad=False`. Inner folds select the treatment threshold, not the training epoch. Every model therefore trains for 80 epochs. The ensemble uses base seeds 17, 43, and 89, with deterministic offsets for the outer fold, inner fold, and depth defined as follows. For base seed b, outer fold $o \in \{0, \ldots, 5\}$, encoder depth $d \in \{1, 8\}$, and global inner fold ID $j \in \{0, \ldots, 5\} \setminus \{o\}$, the training seed is $b + 1{,}000{,}003\, o + 10{,}007(j + 1) + 100{,}003\, d$; the fit used to score the six scenes in the outer test omits the $10{,}007(j + 1)$ term, and the fit on all development scenes uses $b + 100{,}003\, d$. At each depth, raw head outputs are averaged over seeds; the sigmoid is then applied only to the three averaged binary logits, while the two gain outputs remain on their regression scale before treatment selection. These offsets uniquely specify the RNG seed for every fit. We treat the three seeds as one initialization ensemble and report no variance estimate across them. One pair of heads for blocks 1 and 8 has 269,884 learned parameters; the ensemble of three seeds has 809,652 learned parameters and shares one fixed feature extractor with 182,529 parameters, for a total of 992,181 stored selector parameters, excluding the frozen CroCo backbone. Appendix B.2 reports every subset containing one, two, or three seeds without choosing among them after seeing the results.

A.3 Nested cross-validation over scenes

Table 4: Fixed corruption families used for threshold constraints and reporting. Clean is evaluated separately. Family labels are not selector inputs.

Family	Corruption types
Color	brightness, contrast, saturate
Blur	defocus blur, Gaussian blur, glass blur, motion blur, zoom blur
Noise	Gaussian noise, impulse noise, shot noise, speckle noise
Quality	pixelation, JPEG compression, elastic transform
Weather	fog, frost, rain, snow, spatter

Testing uses scene-disjoint folds. The 36 scenes are split into six disjoint test groups of six scenes. For each outer fold, its six test scenes are excluded from both training the prediction heads and selecting the threshold. On the remaining 30 scenes, five models each train on 24 scenes and predict the other six. Combining these predictions gives one score for every example used to choose the threshold; each score comes from a prediction head that did not train on that example's scene.

We test up to 401 thresholds, chosen to produce evenly spaced numbers of changed examples from zero to the total number of predictions; rounding duplicates are removed. The no-change option uses $\tau = +\infty$; every other threshold is the score of the last example that would be changed. We retain thresholds with clean gain $\geq -0.2\%$, gain in every corruption family ≥ 0, and gain in the worst scene $\geq -0.5\%$. Ties are resolved by C_5, then mean gain across corruption types, and finally $\text{Harm}_{.001}$(all), the number of changed corrupted examples whose Abs rises by more than 0.001 px divided by all corrupted examples. We fit the three-seed ensemble on all 30 training scenes and evaluate it once on the six excluded scenes at the selected threshold. Repeating the procedure gives one threshold per test group. Color gain is -0.02% after combining the six tests, so that training-side requirement did not carry over to the excluded scenes; we left the rule unchanged after testing.

After the tests on all six excluded scene groups, the six ensembles have produced one score for every development example: each ensemble was trained on 30 scenes and scored its six excluded scenes. Pooling the resulting 8,784 scores and applying the same constraints and tie-breaking rule selects the deployment threshold, $\tau = 0.6998358965$. Six prediction heads (two depths times three seeds) are then trained on all 36 scenes. We freeze those heads, the feature extractor, and the threshold before inspecting scene 0022 or running official inference. A subsequent run on the development data serves only as a consistency check and contributes no new excluded-scene result.

To measure sensitivity to scene composition, we draw 36 scenes with replacement 5,000 times. Each draw carries all frames, corruptions, predictions, and treatment choices from its sampled scenes, and paired comparisons use the same scene multiplicities. We report the 2.5th and 97.5th percentiles of the recomputed metric estimates. Prediction heads, thresholds, and treatment decisions remain fixed throughout, so these ranges exclude retraining and model-selection variation.

Table 5: Scene roles in nested cross-validation and final deployment. A dash means the stage does not use scenes for that purpose.

Stage	Train prediction heads	Choose threshold	Reported use
Models for inner folds	24 of the other 30 scenes	Predictions on the other 6; combined over all 30	Select one threshold for each outer fold
Models for each outer test fold	All other 30 scenes	Fixed from inner folds	Score excluded scenes; collectively yield one score for each development example from a model not trained on its scene
Selecting deployment threshold	—	All 8,784 scores described above	Select one final threshold
Deployment heads	All 36 development scenes	Fixed final threshold	Final system construction; not an independent test result
Checks after settings are fixed	—	—	Scene 0022 (outside the 36 development scenes) and official images

A.4 Metric aggregation

For corrupted examples $\mathcal{C}$ and subsets $\mathcal{C}_c$ for each corruption type, define the relative reduction in Abs (positive means lower error) as

$$R(S) = 100\,\frac{\sum_{i\in S} g_{i,\pi(i)}}{\sum_{i\in S} L_{i,\mathrm{id}}}, \qquad R_{\mathrm{overall}} = R(\mathcal{C}), \qquad R_{\text{type avg.}} = \frac{1}{20}\sum_{c=1}^{20} R(\mathcal{C}_c).$$

The crop results sum *mean* errors over valid pixels, giving each example equal weight. They pool the three severities within each corruption type before taking that type's relative gain; they do not average the relative percentage for each example. The evaluation on five crops instead sums absolute error over valid pixels in the five crops inferred separately. Equivalently, replace $g_i, L_{i,\mathrm{id}}$ above by $n_i g_i, n_i L_{i,\mathrm{id}}$, where n_i counts scored valid pixels including repeated overlap. Gains for each corruption family and scene use the same ratio; worst scene is the minimum scene gain. Clean gains are computed separately. For a given rule, the fraction modified is the fraction of corrupted examples assigned a treatment other than unchanged. Harm$_{.001}$ and harm$_{.05}$ count changed corrupted examples with $g < -0.001$ and $g < -0.05$ px, respectively; "all" divides by all corrupted examples and "modified" by all changed corrupted examples under that rule. The percentage of improvement under the upper bound likewise excludes clean examples and uses the denominator specified for each comparison.

A.5 Submission inference at full resolution

After the five-crop vote in Section 3, one treatment is applied to the complete stereo pair. On a native 1080×1920 pair, the five 352×704 selector crops have top-left coordinates $(y, x) = (347, 608), (104, 135), (104, 1081), (624, 135), (624, 1081)$ for the center, upper-left, upper-right, lower-left, and lower-right crops, respectively. Each crop uses the half-open region $[y, y + 352) \times [x, x + 704)$. The voting center is snapped to the official inference grid and therefore differs from the center at $(364, 608)$ used in the local evaluation. Their union contains 1,154,032 of 2,073,600 pixels (55.653549%); scoring the crops separately counts pixels in overlaps once per crop. The released CroCo-Stereo tiler [4] runs aligned 352×704 windows over the native images without resizing. For axis length L and window size w, it uses $n = 1 + \lceil (L - w)/(0.1w) \rceil$ starts, obtained by rounding n evenly spaced positions from 0 to $L - w$. A target overlap of 0.9 therefore gives $22 \times 19 = 418$ windows for each reference view, covering the full image.

For tile t, disparity $d_t(p)$ and raw auxiliary channel $c_t(p)$ at pixel p, the checkpoint's `conf_expsigmoid_15_3` rule is

$$w_t(p) = \exp\left[-30\left(\sigma(c_t(p)/3) - \tfrac{1}{2}\right)\right], \qquad d(p) = \frac{\sum_{t:p\in t} w_t(p) d_t(p)}{10^{-16} + \sum_{t:p\in t} w_t(p)},$$

where σ is the logistic sigmoid. Overlapping predictions are accumulated at their original pixel coordinates. Inference with the right image as reference swaps the views and horizontally flips both inputs; the resulting disparity is flipped back. Both references share the selected treatment. This path generates the clean and corrupted submission maps used to compute RbS 0.1456; the five crops select the treatment but are not themselves stitched into the output.

B Additional results

B.1 Local baselines and upper bounds

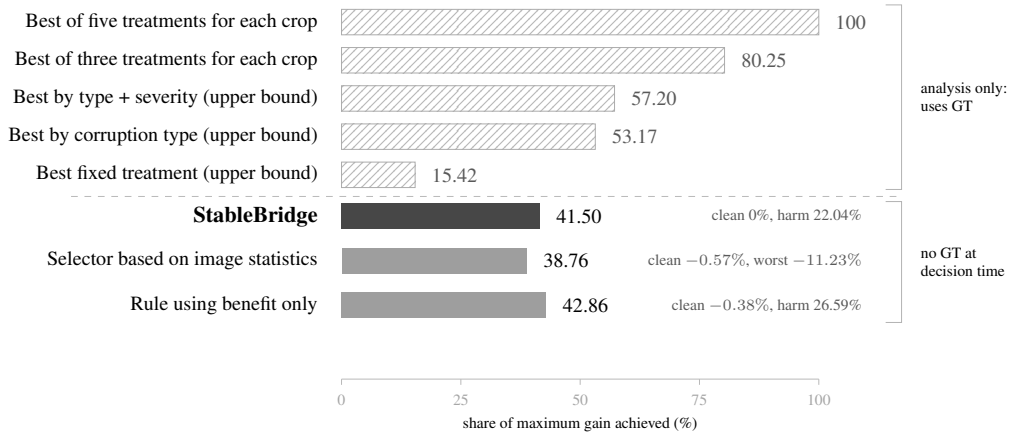


Figure 3: The bars show how much of the five-treatment GT upper bound each rule captures. Hatched bars are analysis upper bounds that use GT; solid bars use no GT when choosing a treatment. StableBridge and the benefit-only rule use the same block-1 proposal and, within each outer fold, change the same total number of clean and corrupted examples. Annotations report clean gain and harm rate.

Local disagreement between clean and corrupted predictions. For the results in Section 5, we reuse the saved predictions for all 8,640 corrupted crops. For each crop, we compare CroCo's prediction on the clean image with predictions on the corrupted image from CroCo alone and from the StableBridge model that excluded that scene. Computing prediction disagreement uses no GT; the approximate local corruption generator can nevertheless use Spring disparity to create effects that depend on depth. Scoring the complete crop differs from official test sampling.

Against crop GT, StableBridge also reduces the fraction of pixels above 1 px error from 26.373% to 25.391% and D1 from 11.629% to 11.204%. The relative reductions are 3.72% and 3.65%, respectively. These are GT error rates, whereas Δ compares predictions.

The scalar noise score averages, over both views, the standard deviation of the pixelwise absolute difference between an image and a copy blurred with a Gaussian kernel of $\sigma = 0.7$. Table 6 places the learned selectors, scalar baselines, and GT upper bounds on the common C_5 scale.

The 53.2% and 57.2% upper bounds select the best of five treatments separately for each corruption type or type–severity group, respectively, using all labeled corrupted examples rather than estimating performance on excluded scenes:

$$a_c^{\text{best}} = \arg\max_{a \in \mathcal{A}_5} \sum_{i: c_i = c} g_{ia}, \qquad a_{c,s}^{\text{best}} = \arg\max_{a \in \mathcal{A}_5} \sum_{i: c_i = c,\, \text{sev}_i = s} g_{ia}.$$

Here sev_i is the corruption severity. These are empirical upper bounds for treatments that remain constant within each group. For each test group, a threshold learned from the other 30 scenes chooses denoising or the identity treatment; the requirements for clean images, corruption families, and the worst scene cause it to change only 0.81% of corrupted examples. Removing those requirements while selecting the threshold on the same 30 scenes gives "Scalar noise score, threshold from training scenes." Thresholds are compared by C_5, then mean gain across corruption types, and finally the

fraction of changed corrupted examples whose Abs rises by more than 0.001 px. This scalar baseline is separate from the benefit-only learned rule. An analysis-only threshold chosen from all labeled outcomes reaches $C_5 = 16.57\%$ while modifying 44.19% of corrupted examples; it raises clean Abs by 2.03% and worst-scene Abs by 14.27%. Because the threshold sees those outcomes, we use it only as an upper bound for the scalar rule on this labeled set.

Table 6: Local crop comparisons; C_5 is defined in Eq. 4, and positive gain means lower error. Harm uses the 0.001 px cutoff.

Rule	Mean by type	C_5	Modified	Clean	Harm (all)	Harm (modified)
Unchanged CroCo	0.00%	0.0%	0.0%	baseline	0.0%	—
One fixed treatment for all corrupted examples (chosen with GT)	+2.28%	15.4%	100.0%	—	54.3%	54.3%
Scalar noise score, constrained threshold	+0.14%	1.6%	0.8%	0.00%	0.3%	37.1%
Scalar noise score, threshold from training scenes	+0.86%	2.9%	72.2%	−2.42%	37.5%	51.9%
Scalar noise score, threshold chosen using GT	+1.77%	16.6%	44.2%	−2.03%	21.7%	49.0%
Selector based on image statistics	+4.10%	38.8%	75.0%	−0.57%	28.1%	37.4%
StableBridge, three seeds	**+4.29%**	**41.5%**	30.9%	**0.00%**	6.8%	22.0%
Best by corruption type (uses GT)	+5.70%	53.2%	—	—	—	—
Best by type + severity (uses GT)	—	57.2%	—	—	—	—
Best of five treatments for each crop (uses GT)	+10.07%	100.0%	—	—	0.0%	0.0%

Mean by type, C_5, modified, and harm use the same 8,640 rows from corrupted center crops; clean uses 144 separate rows from clean crops. Modified is the percentage assigned a non-identity treatment. Harm (all) divides changed rows whose Abs increases by more than 0.001 px by 8,640; harm (modified) divides the same rows by that rule's changed corrupted rows. A dash denotes an unavailable or undefined quantity, not zero. Rows differ in inputs, fitting, treatment sets, and modification rates; Table 13 gives the matched comparison for decision-time use of the harm score.

The baseline based on image statistics uses 76 statistics to assign weights to the corruption types without GT. It then uses those weights to average the gains for each treatment measured on the fitting scenes and selects the treatment with the largest estimated gain. These weights combine results across corruption types and carry no probabilistic interpretation. The baseline may choose all five treatments in the analysis set and therefore has access to more potential improvement than StableBridge with three treatments. Its inputs, fitting, and modification rate also differ, so the C_5 gap mixes several design choices.

B.2 Development frames and ensemble checks

We repeated the complete procedure with six folds using eight rather than four development frames per scene. After sorting the filenames in each scene of length N, the eight positions are $\lfloor jN/10 \rfloor$ for $j \in \{1, 2, 3, 4, 6, 7, 8, 9\}$; the original four positions are the subset $j \in \{2, 4, 6, 8\}$. The development set with eight frames has 288 stereo pairs and 17,568 examples, including clean and corrupted images. All 216 prediction-head instances were retrained with seeds 17, 43, and 89; for each outer fold, its threshold was selected again from predictions made by inner-fold models that had not trained on the predicted scene, using the eight-frame data, the same set of up to 401 thresholds, and the same three requirements. Both settings are evaluated on the original four frames per scene. This is a full-procedure comparison: head fitting and threshold selection both change, while the evaluation frames remain fixed.

Table 7: Four- versus eight-frame development sets, evaluated on the same four frames per scene. Each setting refits its heads and reselects its thresholds. Harm counts changed corrupted examples whose Abs rises by more than 0.001 px.

Development frames/scene	Mean gain by type	Modified	Harm/modified
4	4.292%	30.880%	22.039%
8	3.432%	18.704%	19.121%

From the unrounded values, using eight development frames lowers mean gain across corruption types by 0.859 percentage points. Because both head fitting and threshold selection are repeated, the experiment does not isolate sample count from the newly selected threshold.

To assess ensemble size, we evaluated all seven nonempty subsets of the three seeds fixed before evaluation. Every subset uses the same CroCo features, scene folds, prediction-head recipe, and inner-fold threshold rule. We report all subsets and retain the ensemble chosen before these results were available.

Table 8: Results using one, two, or three seeds. Overall and mean by type are relative reductions in Abs on corrupted examples; clean is a relative Abs gain, with positive values meaning lower error; harm is the percentage of changed corrupted examples whose Abs increases by more than 0.001 px.

Seeds	Overall	Mean by type	C_5	Modified	Harm	Clean
17	5.269%	4.444%	43.051%	29.745%	21.868%	−0.157%
43	5.213%	4.370%	42.594%	30.972%	22.683%	0.000%
89	4.880%	4.170%	39.878%	30.104%	22.030%	0.000%
17, 43	5.087%	4.268%	41.568%	30.347%	22.044%	0.000%
17, 89	5.070%	4.281%	41.426%	29.757%	21.937%	0.000%
43, 89	5.097%	4.278%	41.647%	30.845%	22.176%	0.000%
17, 43, 89	5.079%	4.292%	41.501%	30.880%	22.039%	0.000%

Scene resampling shows no consistent advantage for a smaller seed subset in overall gain or C_5. Seed 17 is highest when all scenes are used once, but choosing it now would make the evaluation part of model selection. We therefore keep the three-seed ensemble fixed before evaluation. Because scene resampling holds the trained heads fixed, it excludes variation from training fresh seeds.

B.3 Check of a fixed learned restorer

We evaluated the released SwinIR model for removing JPEG artifacts from color images, trained for JPEG quality 10 [18]. Its weights are frozen, with no fine-tuning on Spring. The model processes both views at full resolution independently with shared weights and preserves their image grid. We tested its full output ($\alpha = 1$) and a half blend with the input ($\alpha = 0.5$); both settings were fixed before evaluation. This deterministic experiment fits no selector; we resample complete scenes 5,000 times only to show sensitivity to scene composition. The evaluation set contains 1,872 cases, each a stereo pair under one condition, from 36 scenes: clean images, all three JPEG severities, and three severities each of Gaussian noise, defocus blur, and rain to check for unintended harm outside JPEG.

Table 9: The fixed SwinIR restorer on JPEG severities 1–3. Positive gain means lower CroCo disparity error than unchanged input. The last column gives paired values from the scene resampling in Appendix A.3. The half blend probes sensitivity to restoration strength; the full output remains the prespecified candidate for adding a fourth treatment.

Input to CroCo	Abs (px)	Gain (px)	Range from resampled scenes (px)
Unchanged	4.884	0.000	0.000 to 0.000
Mild denoising	4.586	+0.297	+0.094 to +0.583
Mild sharpening	5.657	−0.773	−1.276 to −0.353
SwinIR, $\alpha = 1$	4.702	+0.182	−0.555 to +0.887
SwinIR, $\alpha = 0.5$	4.860	+0.024	−0.555 to +0.620

Using all scenes once, the full SwinIR output improves JPEG images by 3.72%, but estimates from resampled scenes include both improvement and degradation. It also increases Abs on clean images from 0.684 to 1.066 px and raises Abs across the nine conditions other than JPEG used to check unintended harm from 2.107 to 3.100 px.

Before running this experiment, we required the reported range from resampling scenes for the gain of the full SwinIR output on JPEG images to remain above zero before continuing to a selector with four treatments. All 1,872 cases were processed, but the range extended below zero, so we stopped after evaluating this fixed SwinIR model. Other learned restorers and a separately trained four-treatment selector remain untested.

B.4 One treatment for each pair, scored on five crops

Table 10 compares the five-crop vote defined in Section 3 with a decision from the center crop and separate decisions for each crop, as well as upper bounds using GT.

Table 10: Pair-level treatment choices at the middle severity, scored by separate inference on five crops. "Decision from center crop" chooses from the center crop but still scores all five. Abs is mean absolute disparity error against GT on corrupted crops; Clean, Mean by type, and Worst scene are relative gains over CroCo alone.

Rule	Clean	Abs (px)	Mean by type	Positive	Worst scene
CroCo alone	0%	3.594	0%	—	—
Five-crop vote	**0%**	3.423	**+3.465%**	10/20	−4.569%
Decision from center crop	−0.026%	3.419	+3.561%	12/20	−4.602%
Separate treatment for each crop	−0.027%	3.409	+3.769%	15/20	−2.355%
Best treatment for each pair (uses GT)	+0.132%	3.218	+7.666%	20/20	+4.165%
Best treatment for each crop (uses GT)	+0.682%	3.125	+9.435%	20/20	+5.872%

"Positive" counts corruption types with positive mean gain. Rows using GT are upper bounds for analysis.

The five-crop vote reduces corrupted GT Abs by 4.778% from the unrounded values. Clean Abs stays at 1.103 px. The center-crop rule and five-crop vote retain their original thresholds; on the same unrounded basis, the vote's gain is 0.094 percentage points lower than the center-crop result. The vote changes 29.792% of corrupted cases; 5.556% of all corrupted cases, or 18.648% of changed cases, increase Abs by more than 0.001 px. We did not retune either rule to equalize only the number of changed corrupted examples.

B.5 Results by corruption type

Figure 4 shows the same comparison at the middle severity between the decision from the center crop and the five-crop vote, reported by corruption type in Table 11.

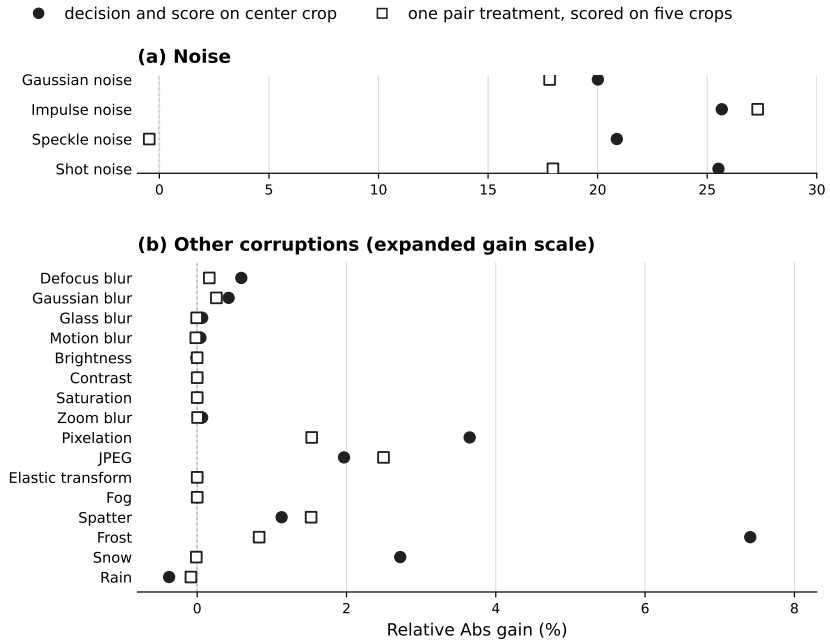


Figure 4: Per-pair treatment performance across five crops for the same 144 pairs at the middle severity. Circles show the center-crop decision scored on that crop; squares show the five-crop vote scored across all five crops. For speckle noise, denoising wins 132/144 votes and improves the center by 18.16%, but the four quadrants regress by 2.64%, giving −0.462% overall.

Table 11: Relative GT Abs gain (%) on corrupted crops at the middle severity, by corruption. Positive values mean lower error. "Center" uses the decision from the center crop and scores that crop; "Pair vote" applies one selected treatment to the pair, then infers and scores five crops separately.

Corruption	Center	Pair vote	Corruption	Center	Pair vote
Gaussian noise	20.011	17.809	Impulse noise	25.665	27.303
Speckle noise	20.875	−0.462	Shot noise	25.514	17.960
Defocus blur	0.591	0.162	Gaussian blur	0.421	0.257
Glass blur	0.067	−0.007	Motion blur	0.044	−0.018
Brightness	−0.011	0.000	Contrast	0.000	0.000
Saturation	0.000	0.000	Zoom blur	0.066	0.005
Pixelation	3.650	1.534	JPEG	1.967	2.496
Elastic transform	0.000	0.000	Fog	0.000	0.000
Spatter	1.131	1.526	Frost	7.409	0.830
Snow	2.720	−0.012	Rain	−0.376	−0.084

The mean gain across corruption types on center crops at the middle severity, computed from Table 11, is 5.487%; the main 4.292% result instead pools severities 1–3 within each corruption type before averaging the 20 gains, one for each type.

Let r_c be the reported relative Abs gain for corruption type c and $\mathcal{N}$ the four noise types. From the rounded values in Table 11, the mean gain across corruption types in the evaluation on five crops is approximately 3.465%, and the mean for the other 16 types is

$$\frac{1}{16}\sum_{c\notin\mathcal{N}} r_c \approx 0.418\%, \qquad \frac{\sum_{c\in\mathcal{N}} r_c}{\sum_{c=1}^{20} r_c} \approx 90.3\%.$$

Computed directly from the rounded entries, the second quantity gives noise's share of equal-weight, type-level net gain. It includes negative gains and is separate from the scene-resampling analysis.

Saved speckle outputs show that a treatment chosen from the crops may fail elsewhere in the image. The five-crop vote selects denoising on 132/144 pairs. With denoising applied to the complete pair, the center crop improves by 18.16%, but error in the four quadrant crops rises by 2.64%, producing -0.462% over all five crops. The best treatment for each pair, selected with GT for this analysis, still gains 11.18%; on 24 pairs StableBridge selects denoising while the best choice is unchanged input. Speckle therefore shows both failure to transfer across the image and selection of a treatment other than the one with lowest error. This diagnosis stops at the five scored crops, before the complete-map stitching in Section 3 and Appendix A.5.

For rain, gain changes from $+5.414\%$ when all three severities on center crops are combined to -0.376% on center crops at the middle severity. Scene 0022, outside the development set, supplies a concrete example: denoising has a measured $+9.195$ px gain but is rejected at the deployed threshold.

B.6 Official RobustSpring submission

The six trained prediction heads, fixed feature extractor, and threshold 0.6998358965 were frozen before official Spring and RobustSpring inference. The five-crop vote leaves all 1,000 official clean pairs unchanged and selects unchanged/denoised/sharpened input 14,351/3,650/1,999 times on the 20,000 corrupted pairs. Because the matched CroCo-only server run was not completed, this subsection reports the submitted system end to end.

The corresponding official 1px and D1 disagreement rates are 22.529% and 4.427%. Clean Spring Abs is 0.471 px, hence the leaderboard proxy is $0.471 + 2.386 = 2.857$ px and Eq. 5 gives RbS 0.1456. The mean of the rounded entries for each corruption in Table 12 is 2.38635 px, consistent with the reported total.

The three clean Abs values use different pixel sets: 0.684 is the comparison on the center crop, 1.103 pools valid pixels from the five scored crops, and 0.471 is the official server result over complete images. Only comparisons within the same evaluation setting are meaningful.

Table 12: Official disagreement Δ (px) between predictions on clean and corrupted images for the frozen StableBridge submission. These values come from the test server and are not errors on corrupted images against ground truth.

Corruption	Δ_{Abs}	Corruption	Δ_{Abs}
Brightness	0.303	Contrast	0.375
Defocus blur	0.326	Elastic transform	1.682
Fog	0.555	Frost	7.177
Gaussian blur	0.420	Gaussian noise	1.652
Glass blur	0.413	Impulse noise	1.637
JPEG compression	4.200	Motion blur	1.341
Pixelate	1.338	Rain	3.936
Saturate	0.207	Shot noise	1.447
Snow	5.310	Spatter	6.836
Speckle noise	1.441	Zoom blur	7.131
Overall	**2.386**		

The largest official disagreements occur for frost, zoom blur, spatter, and snow, while evaluation on five crops shows at most 1.53% gain for those four corruptions. This supports the practical limitation of the current treatment set: it is weak outside noise. We use official disagreement to describe the submitted system and local GT gain to analyze selection, without equating their numerical values.

B.7 Design comparisons and negative results

On the 30 development scenes used to design the architecture, the conservative-gain score from block 1 has a slightly higher Spearman correlation with measured treatment gain than the score from block 8 (0.551 versus 0.538). The block-8 auxiliary score for detecting Abs increases above 0.05 px has higher AUROC than its block-1 counterpart (0.797 versus 0.745). Separate heads at the two depths yield $C_5 = 41.40\%$ and 4.271% mean gain across corruption types, compared with 27.24% and 2.671% for one head receiving both feature vectors. These development results motivated the architecture, but the deployed harm rule uses the 0.001-px target; Table 13 provides the relevant decision-time comparison.

Table 13 compares rules that change the same total number of examples, isolating how the block-8 harm score affects the final decision to apply a treatment or keep the pair unchanged.

Table 13: Rules matched to the same total number of changed examples within each test group. That total includes clean and corrupted examples, so corrupted-only modification rates may differ. Mean by type and Clean are relative Abs gains; $\mathrm{Harm}_{.001}$ and $\mathrm{Harm}_{.05}$ count changed corrupted examples whose Abs rises by more than 0.001 px and 0.05 px.

Selection score	C_5	Mean by type	Modified	$\mathrm{Harm}_{.001}$	$\mathrm{Harm}_{.05}$	Clean
Conservative gain score only	24.41%	+2.258%	30.46%	36.21%	9.19%	−0.530%
Rule using benefit only	42.86%	+4.369%	30.78%	26.59%	12.30%	−0.376%
Benefit minus harm	**41.50%**	**+4.292%**	30.88%	**22.04%**	**11.02%**	0.000%

For the matched rows, the block-1 proposal stays fixed and only the final score changes. Both rows use heads trained with the harm objective, so the comparison isolates the decision-time effect of the saved block-8 score.

The C_5 difference relative to the benefit-only rule is -1.36 percentage points (paired 95% scene-bootstrap CI $[-3.77, 1.45]$), providing no clear evidence of a utility advantage for either rule. This interval uses the 5,000 paired scene resamples in Appendix A.3, with fitted models, thresholds, and decisions fixed.

B.8 Decision examples checked after settings were fixed

Frost illustrates accepted harm: sharpening raises Abs from 5.5784 to 10.7962 px, while denoising would yield 3.8106 px. Rain shows a missed gain at the deployed threshold. These cases come from

one frame and illustrate individual decisions; they were not used to retrain a head or change the threshold.

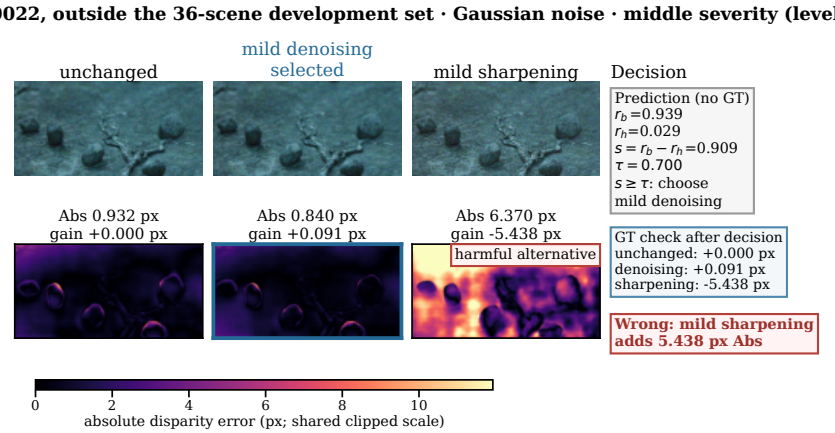


Figure 5: A helpful selection and a harmful alternative on scene 0022, outside the 36 development scenes. Gaussian noise is applied at the middle severity. Top: candidate crops; bottom: disparity-error maps on one shared clipped color scale; right: treatment selection and local outcome. Denoising is applied ($s = 0.909 \geq \tau = 0.700$), while unselected sharpening raises Abs by 5.438 px. Benefit, harm, and difference scores in the panel are rounded independently from full-precision values. Error maps and displayed Abs values are computed against ground-truth disparity; Appendix A.3 gives the exact threshold and freeze point.

Table 14: Middle-severity crop examples from scene 0022, frame 0001, using the deployment threshold displayed as $\tau = 0.700$. The scene lies outside the 36 development scenes, and outcomes were inspected after the decision was fixed. "Choice from block 1" precedes the harm check; DN/SH/ID mean denoising, sharpening, and unchanged input. Scores r_b and r_h come from the weighted classifiers. Gain is the Abs change that the block-1 proposal would produce even when the final decision is ID; positive values indicate lower error.

Corruption	Choice from block 1	r_b	r_h	$r_b - r_h$	Decision	Gain (px)
Brightness	DN	.575	.657	−.082	ID	−.0049
Contrast	DN	.414	.646	−.232	ID	−.0235
Gaussian blur	SH	.833	.152	.681	ID	+.0021
Defocus blur	DN	.404	.693	−.289	ID	−.0101
JPEG	DN	.885	.078	.807	DN	+1.5722
Pixelation	DN	.835	.028	.808	DN	+.3316
Shot noise	DN	.935	.026	.909	DN	+.0313
Frost	SH	.885	.042	.843	SH	−5.2178
Rain	DN	.816	.345	.471	ID	+9.1951
Snow	DN	.700	.332	.368	ID	+1.5045

C Implementation, assets, and compute

Experiments used Python 3.12.14, PyTorch 2.6.0+cu124, NumPy 2.4.4, OpenCV 4.13.0, and Pillow 12.2.0. All reported timings and resource measurements used NVIDIA RTX 3090 GPUs. On four GPUs, generating CroCo outputs for five treatments took 69 minutes and extracting features took 30 minutes; development comparisons and head training took 70 minutes on one GPU. The timing check uses official tiling with 0.9 overlap, batch size 1, one pair for warm-up, and eight measured pairs; image decoding is excluded. Median times for CroCo, the selector, and both combined are 89.29, 1.22, and 90.58 seconds for each pair, respectively, so the selector adds about 1.4% runtime in this BF16 configuration. Peak memory is 2.59 GiB for the selector and 2.12 GiB for a complete CroCo forward pass; the two peaks were measured separately and do not represent simultaneous usage. Input selection and the timed CroCo run both use 16-bit bfloat16 (BF16). The submitted

configuration uses BF16 for selector features and 32-bit floating point (FP32) for final disparity inference; conversion to the required `float32` storage type does not change compute precision.

Spring/RobustSpring are licensed under CC BY 4.0, CroCo under CC BY-NC-SA 4.0, and the development kit under Apache 2.0; the study introduces no human subject data or scraped personal data.

Producing a prediction for every scene from models not trained on that scene requires fitting 216 prediction heads; the deployed model stores six. The reported $C_5 = 41.5\%$ uses the tests on excluded scenes; the deployed model's $C_5 = 43.1\%$ on all development scenes is only a consistency check.

D Broader impacts

Selective treatment may improve reliability but can increase error and latency; it is not a safety certificate. Deployment requires validation on the target sensor, monitoring, and an independent fallback. Stereo can support surveillance; privacy and governance are outside our scope.